

and computer vision techniques to estimate the poses of multiple people in real-time (<https://github.com/CMU-Perceptual-Computing-Lab/openpose>).

- **TensorFlow Object Detection API:** This is an open source library for object detection and classification. It uses TensorFlow and OpenCV for training and inference of deep neural networks for object detection in real-time scenarios (https://github.com/tensorflow/models/tree/master/research/object_detection).
- **DLib:** DLib is a modern C++ toolkit for building computer vision applications. It includes a wide range of algorithms and tools for object detection, face recognition, and more (<https://github.com/davisking/dlib>).
- **OpenCVSharp:** This is a .NET wrapper for the OpenCV library. It allows developers to use OpenCV in their .NET applications.
- **Emgu CV:** A cross-platform .NET wrapper for the OpenCV library, it provides access to OpenCV functions from .NET languages such as C#, VB.NET, and F#.
- **OpenFace:** OpenFace is an open source facial behaviour analysis toolkit that uses OpenCV for facial landmark detection and tracking.
- **Mahotas:** A Python library for image processing and computer vision, it uses OpenCV for basic image processing operations.
- **scikit-image:** This Python library for image processing and computer vision uses OpenCV for some image processing functions.

These OpenCV-based open source libraries provide developers with a range of tools and functionalities for building AI and real-time computer vision applications.

Leading platforms for computer vision

If you are looking for similar options as the OpenCV platform, we have

you covered. Here are some popular programming platforms and libraries for computer vision with their respective URLs.

- **TensorFlow:** TensorFlow is an open source platform for developing and deploying machine learning models. It includes a module for computer vision tasks such as image recognition, object detection, and segmentation (<https://www.tensorflow.org/>).
- **PyTorch:** PyTorch is an open source machine learning framework that offers a range of tools and libraries for building and deploying deep learning models. It has a module for computer vision tasks such as image classification, object detection, and segmentation (<https://pytorch.org/>).
- **Keras:** Keras is a high level neural networks API, written in Python and capable of running on top of TensorFlow, CNTK, or Theano. It provides a simple and intuitive way to build and train deep learning models for computer vision tasks such as image recognition and segmentation (<https://keras.io/>).
- **MATLAB:** MATLAB is a programming language and numerical computing environment widely used in engineering and scientific applications. It includes a comprehensive toolbox for computer vision tasks such as image processing, computer vision system design, and machine learning (<https://www.mathworks.com/products/computer-vision.html>).
- **SimpleCV:** SimpleCV is an open source framework for building computer vision applications in

Python. It provides a high level interface for common CV tasks such as image processing, feature extraction, and object recognition (<https://simplecv.org/>).

- **TorchVision:** TorchVision is a computer vision library that is part of the PyTorch ecosystem. It provides tools and models for image and video classification, object detection, and segmentation (<https://pytorch.org/vision/>).
- **ImageJ:** ImageJ is open source image processing and analysis software written in Java. It includes a range of tools for image processing, segmentation, and feature extraction (<https://imagej.net/>).

These platforms and libraries provide developers with a range of tools and functionalities to build and deploy computer vision applications.

Final thoughts

It's no secret that AI is the future. The synergy between computer vision and IoT can solve some of the world's most pressing issues. But as exciting as that sounds, the technology has remained quite exclusive to those who have the budget to back it up. That's where OpenCV comes in.

By democratising AI, platforms such as OpenCV are redefining scenarios that have traditionally been out of reach for smaller businesses. This shift not only promotes innovation but also opens doors for new talent and creates an opportunity for previously underserved markets to thrive.

Having access to powerful technology shouldn't be exclusive. OpenCV helps level the playing field for everyone. **END** 🐧

By: Dr Gaurav Kumar and Amit Doegar

Dr Gaurav Kumar is associated with various academic and research institutes for delivering expert lectures and conducting technical workshops on the latest technologies and tools.

Amit Doegar is an associate professor in the Department of Computer Science and Engineering, National Institute of Technical Teachers Training & Research (NITTTR), Chandigarh.